Impact of Enhanced Seasonal Malaria Chemoprevention on Haemoglobin Trajectories in Children

Samuel AGYEKUM — Philomene CYUBAHIRO — Djadida UWITUZE —Rabecca KATING'U — Foumabn Marah Khadij NOURIA

Abstract

Malaria is a life-threatening disease caused by parasites transmitted to people through the bites of infected female Anopheles mosquitoes.It is a leading cause of severe childhood anaemia across sub-Saharan Africa. We analyzed a six-month longitudinal randomized controlled trial (RCT) to assess whether an Enhanced Seasonal Malaria Chemoprevention (SMC) protocol, which adds high-dose iron and nutritional supplements to standard antimalarials, better protects hemoglobin levels compared to Standard SMC alone. The trial enrolled 300 children with measurements taken at baseline and at months 2, 4, and 6. We fitted Linear Mixed Models (LMM) to handle correlated repeated measures. The Intraclass Correlation Coefficient confirmed that most haemoglobin variation arose from stable differences between children. A likelihood ratio test favoured a model with both random intercept and random slope. The treatment by time interaction was highly significant, indicating that children on Enhanced SMC gained an extra 0.205 g/dL each month compared to those receiving Standard SMC. Over six months, this amounts to a clinically meaningful advantage of approximately 1.23 g/dL. We recommend that the Ministry of Health consider expanding the Enhanced SMC program to improve childhood anemia prevention.

Keywords: seasonal malaria chemoprevention, haemoglobin, iron supplementation, linear mixed models, longitudinal RCT, childhood anaemia.

Introduction

Malaria caused by *Plasmodium falciparum* kills hundreds of thousands of children every year in sub Saharan Africa [5]. The parasite destroys red blood cells and suppresses bone marrow, causing hemoglobin to drop until children fall into severe anaemia. Even children who survive repeated infections can suffer lasting damage to their brain development and physical growth [1]. Seasonal Malaria Chemoprevention (SMC) gives young children preventive antimalarial drugs during the rainy season and can cut malaria episodes by up to 75% [2]. Yet many children still become anaemic because SMC treats the parasite but not the underlying iron deficiency. A Cochrane review of 35 trials showed that adding iron supplements does not increase malaria risk when proper malaria prevention is in place [3]. This opens the door to combining iron with SMC to protect both against malaria and anaemia at the same time. Each child in this trial was measured four times, so the observations within the same child are correlated. Standard regression ignores this and gives misleading p values. Linear Mixed Models (LMMs) handle the problem by splitting variation into stable between child differences (random effects) and within child changes over time (residuals), producing valid inference for longitudinal data [4].

Study objective: Do children receiving Enhanced SMC show significantly better haemoglobin trajectories over six months than children on Standard SMC?

Methods

Study Design

This was a six-month longitudinal randomized controlled trial that included 300 children,who were measured at four time points: baseline (month 0) and follow-up at months 2, 4, and 6. The children were randomly assigned in a 1:1 ratio to either Standard SMC with control being 155 and Enhanced SMC beging intervention group being 145. The dataset contained 1,200 total observations with no missing values.

Variables

The outcome variable was haemoglobin concentration in g/dL. Fixed effect predictors were treatment group, follow-up month (continuous), age at baseline in months, and ITN use (consistent vs. inconsistent). The critical term was the treatment and month interaction, which tests whether the two groups change at different rates over time.

Model Justification

Each child was measured four times, so observations from the same child are correlated. Ordinary regression assumes independence and would produce standard errors that are too small, giving us false positives. Linear Mixed Models fix this by separating variation into between child differences which is the random effect and within child fluctuations which is the residuals [4]. The ICC from our model was also 0.704, which means that 70% of hemoglobin variation sits between children. An ICC this high confirms that a mixed model is essential. A likelihood ratio test further showed that allowing each child their own rate of change (random slope) fits significantly better than a random intercept alone.

Model Building Strategy

We built the LMM in four steps. First,we fitted an unconditional means model with no predictors, just a random intercept which was to calculate the Intraclass Correlation Coefficient. Next, we added fixed effects starting with month followed by treatment age, ITN and the interaction term, we compared models using AIC for model selection. Thirdly, we compared two random effect models, one with just a random intercept and another with both a random intercept and slope. This comparison was done using a likelihood ratio test and

information criteria. Then finally, the selected model was re-fitted using Restricted Maximum Likelihood for unbiased estimates of the variance.

The final model is:

$$Y_{ij} = \beta_0 + \beta_1 \,\text{Month}_{ij} + \beta_2 \,\text{Trt}_i + \beta_3 \,(\text{Month} \times \text{Trt}) + \beta_5 \,\text{ITN}_i + u_{0i} + u_{1i} \,\text{Month}_{ij} + \varepsilon_{ij}$$

where $u_{0i} \sim \mathcal{N}(0, \sigma^2_{u_0})$, $u_{1i} \sim \mathcal{N}(0, \sigma^2_{u_1})$, and $\varepsilon_{ij} \sim \mathcal{N}(0, \sigma^2)$.

Diagnostics

We checked homoscedasticity (residuals vs. fitted values), normality of observation-level residuals (Q-Q plot), and normality of child-level random effects (Q-Q plots for intercepts and slopes).

Results

Baseline Balance

Table 1 summarises the baseline characteristics. The two groups were broadly comparable at enrolment. Mean age was 34.4 months (Standard) versus 36.9 months (Enhanced) with a p value of 0.094. Baseline haemoglobin were nearly identical with values of 11.98 vs. 12.03 g/dL ($p = 0.668$), confirming that neither group entered the trial with an haemoglobin advantage. Consistent ITN use was higher in the Standard group (63.9% vs. 49.7%; $p = 0.018$), so ITN use was included as a covariate in all models.

Table 1: Baseline characteristics by treatment group.

Characteristic	Standard ($n = 155$)	Enhanced ($n = 145$)	p
Age (months)	34.4 ± 12.4	36.9 ± 13.3	0.094
Haemoglobin (g/dL)	11.98 ± 0.94	12.03 ± 0.96	0.668
ITN use (consistent)	63.9%	49.7%	0.018

Exploratory Data Analysis

Figure 1 shows that from month 2 onwards, the Standard SMC group's haemoglobin declines progressively, approaching the WHO anaemia threshold of 11 g/dL by month 6, while the Enhanced group remains stable.

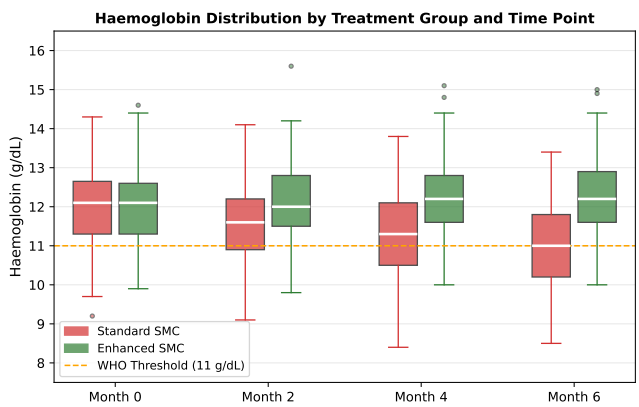

Figure 1: Haemoglobin distributions by group and time point. The orange dashed line marks the WHO anaemia threshold (11 g/dL).

Figure 2 below confirms this divergence where confidence intervals of the two groups no longer overlap by month 4.

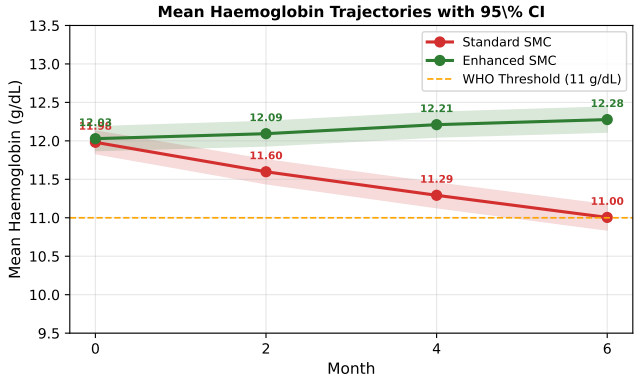

Figure 2: Mean haemoglobin trajectories with 95% CI. Groups start together and progressively diverge.

The spaghetti plot below (Figure 3) shows considerable variability between children in both groups, highlighting the need for random effects in the model.

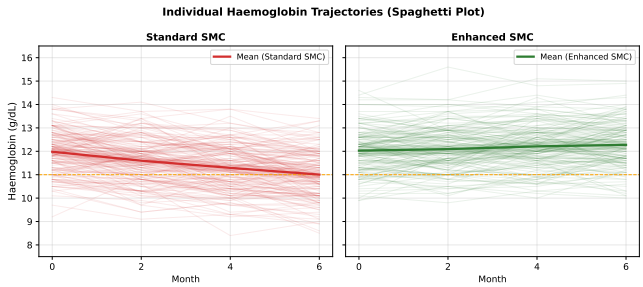

Figure 3: Individual haemoglobin trajectories (thin lines) with group mean ± 95% CI (thick line and shaded band).

Model Building

The unconditional means model yielded a between-child variance of 0.823 and a within-child variance of 0.346, giving ICC = 0.823/(0.823+0.346) = **0.704**. This means **70.4%** of haemoglobin variation is due to stable differences between children, strongly justifying the mixed modelling framework.

Fixed effects: We built the model step by step, adding one predictor at a time and checking whether the AIC has improved. Table (2) below shows the AIC values as each new term was added. The first model included only month as a predictor, and each addition brought the AIC down further. The largest single improvement came from including the treatment and month interaction. When this term was added, the AIC dropped from 2,690.8 to 2,470.5, which is a reduction of 220.3 units. This large drop tells us that the two treatment groups are not just different on average. They are changing at genuinely different rates over the six months, and the model needs this interaction term to describe the data properly.

Table 2: Model comparison as fixed effects are added.

Model	AIC
M1: Month only	2772.7
M2: + Treatment	2733.5
M3: + Age + ITN	2690.8
M4: + Interaction	2470.5

Random effects: The next decision was whether to allow each child have their own rate of change over time or to assume all children change at the same rate.From our Table 3 below the random intercept only model had AIC of 2,470.5 assuming that children start with different hemoglobin levels but follow the same trajectory over time. The model with both the random intercept and the random slope had an AIC of 2,464.2 allowing each child to have their own rate of change in hemoglobin over time. We compared these two models using a likelihood ratio test, which gave a $\chi^2(2)$ of 9.27 and a p value of 0.010. This confirms that allowing each child their own rate of change fits the data significantly better. The result makes biological sense because children differ in how they absorb iron, how often they get bitten by mosquitoes, and how their bodies respond to treatment. Forcing them all onto the same slope would hide important individual variation. The final model was therefore estimated using Restricted Maximum Likelihood (REML) with both a random intercept and a random slope for month.

Table 3: Model comparison as random effects are added

Model	AIC	BIC
Random Intercept	2470.5	2511.2
Random Intercept + Slope	2464.2	2515.1

Model Diagnostics

Figure 4 shows all four diagnostic panels. The residuals vs. fitted plot reveals no systematic pattern and the locally estimated scatterplot smoother (LOESS) stays close to zero, confirming homoscedasticity. All three Q-Q plots (residuals, random intercepts, random slopes) align closely with the reference line, supporting the normality assumptions. Overall, the model assumptions appear to be adequately met.

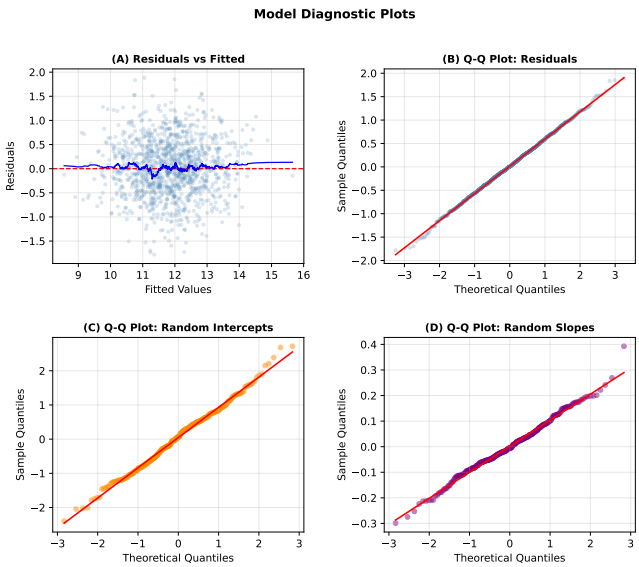

Figure 4: Model diagnostics: (A) Residuals vs Fitted, (B) Q-Q of residuals, (C) Q-Q of random intercepts, (D) Q-Q of random slopes.

Confirmatory Analysis

Table 4 presents the fixed effects from the final REML model. The treatment and month interaction is the central finding of this analysis. Its estimate is $\hat{\beta} = 0.205$ g/dL per month with $p - value < 0.001$. Which means that for every additional month of follow up, children in the Enhanced SMC group gained an extra 0.205 g/dL of haemoglobin compared to children in the Standard group. Over the full six months of the trial, this translates into an advantage of approximately **1.23 g/dL**. The WHO defines childhood anaemia as haemoglobin below 11 g/dL, so a gain of this size is not just statistically significant but clinically meaningful. It is large enough to move many children from the anaemic range into the healthy range, which can have important benefits for brain development, energy levels, and long term growth. The month coefficient which was -0.162 with $p - value < 0.001$ shows what happens to children receiving Standard SMC alone. Their haemoglobin dropped as the rainy season progressed, which is the expected pattern when malaria is controlled but iron deficiency remains. Children who consistently used insecticide treated nets had haemoglobin levels that were 0.430 g/dL higher ($p - value < 0.001$), confirming that bed nets provide additional protection. Older children also had slightly higher haemoglobin, which is a well known pattern in this age group.

Table 4: Fixed effects from the final LMM (REML). The highlighted row is the key result.

Term	Est.	SE	95% CI	p
Intercept	10.965	0.159	[10.653, 11.277]	< 0.001
Month	-0.162	0.010	[-0.181, -0.143]	< 0.001
Treatment (Enh vs Std)	0.078	0.101	[-0.121, 0.276]	0.443
Age at Baseline	0.021	0.004	[0.014, 0.028]	< 0.001
ITN Use (Consistent)	0.430	0.097	[0.239, 0.620]	< 0.001
Month×Treatment	**0.205**	**0.014**	**[0.177, 0.232]**	**< 0.001**

Figure 5 shows the model predicted haemoglobin trajec-

tories for each treatment group, evaluated at the average age and average ITN use in the sample. The dots on the plot are the observed group means at each time point, and the lines are the predictions from our final model. What stands out is that the two lines start at nearly the same level at baseline and then move apart as the months go by. The Standard SMC line slopes downward, showing the steady decline in haemoglobin that malaria driven anaemia causes during the rainy season. The Enhanced SMC line rises slightly, showing that the added iron and nutritional supplements are protecting these children from the decline observed in the Standard SMS group. By month 6 the gap between the two lines reaches 1.23 g/dL, which is marked on the figure. This visual pattern matches the interaction coefficient from the table and confirms that the treatment effect is not a one time jump but a benefit that grows larger with each passing month.

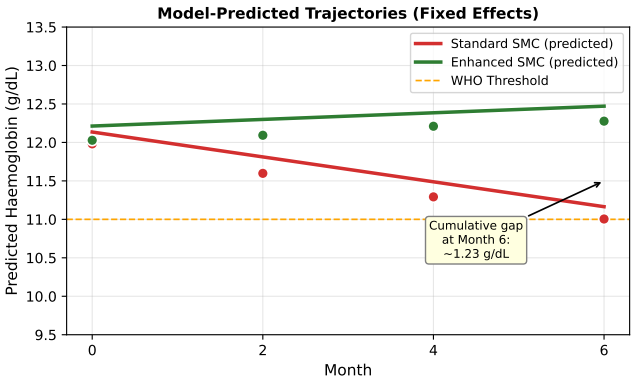

Figure 5: Model-predicted haemoglobin trajectories. Dots show observed group means. The annotation highlights the difference between the two groups at month 6.

Discussion

Main Findings

The Enhanced SMC protocol resulted in significantly better haemoglobin trajectories over six months compared to Standard SMC alone. The treatment-by-time interaction was highly significant, showing that the two groups did not simply differ at single point in time but gradually diverged over the months. This pattern is consistent with the expectation that iron supplementation takes time to rebuild iron stores and stimulate red blood cell production and its full benefit accumulates gradually.

Clinical Significance

A cumulative haemoglobin gain of 1.23 g/dL carries real clinical weight. The WHO anaemia threshold for young children is 11 g/dL. A gain of this size can move many children from anaemic to non-anaemic status, with important benefits for brain development, school performance, and survival. Consistent ITN use was also independently protective ($\hat{\beta} = 0.430$ g/dL, $p < 0.001$), highlighting the value of combining bed-net programs with chemoprevention.

Random Effects

The superiority of both the random intercept and slope model reveals an important biological reality: children do not respond to the rainy season or to the intervention in a uniform way. Some maintained stable haemoglobin throughout, while others declined sharply regardless of treatment. This heterogeneity underscores the need for targeted strategies that identify and prioritise the most vulnerable children.

Limitations

The dataset was simulated; real-world data would introduce missing visits, protocol deviations, and unmeasured confounders such as dietary intake. Follow-up time was modelled linearly, which may not capture any acceleration or plateau in the treatment effect. ITN use was a binary self-report variable, known to overestimate actual bed-net usage. Finally, the six-month window does not tell us whether the haemoglobin advantage persists after the rainy season.

Recommendation

Based on statistical evidence, we recommend that the Ministry of Health consider serious expansion of Enhanced SMC program. The intervention produced a progressive, statistically significant, and clinically meaningful improvement in haemoglobin trajectories. Any national roll out should be accompanied by rigorous monitoring of implementation fidelity, cost-effectiveness, and longer-term haemoglobin outcomes.

References

[1] S. E. Cusick, M. K. Georgieff, and R. Rao. Approaches for reducing the risk of early life iron deficiency induced brain dysfunction in children. *Nutrients*, 10(2):227, 2016.

[2] M. M. Meremikwu, S. Donegan, D. Sinclair, E. Esu, and C. Oringanje. Intermittent preventive treatment for malaria in children living in areas with seasonal transmission. *Cochrane Database of Systematic Reviews*, (2):CD003756, 2012.

[3] A. Neuberger, J. Okebe, D. Yahav, and M. Paul. Oral iron supplements for children in malaria endemic areas. *Cochrane Database of Systematic Reviews*, (2):CD006589, 2016.

[4] G. Verbeke and G. Molenberghs. *Linear Mixed Models for Longitudinal Data*. Springer, New York, 2009.

[5] World Health Organization. World malaria report 2023. Technical report, World Health Organization, Geneva, 2023.

Contributions

1. Samuel AGYEKUM - Report writing + beamer

2.Philomene CYUBAHIRO - Coding + beamer

3.Djadida UWITUZE - Coding

4.Rabecca KATING'U - Report writing

5.Foumabn Marah Khadij NOURIA - Report writing